

Data Transformation

Stage 1



features $\longrightarrow$ values

Compare
Scale

Shape

Normalization

Standardization

sklearn

Scaling (Robust)

function $f(x) = \log x, x^2, \sqrt{x}, x^3$

power transform

Box-Cox

Yeo-Johnson

quantile

Uniform

Normal

$\text{Rank}(x)$

$\Phi^{-1}(\text{Rank}(x))$

Normal CDF

Numeric
features

~~Stage II~~

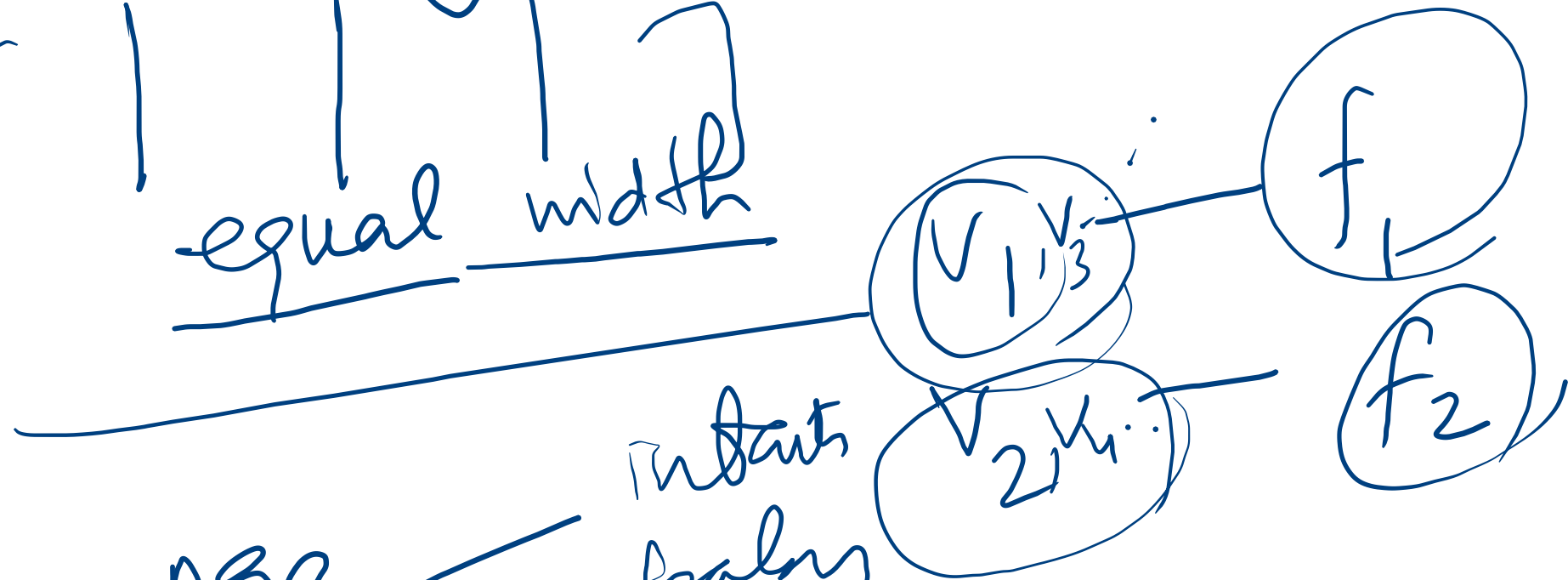
Numeric $\longrightarrow$ Category

$\longrightarrow$ Discretization / Binning

① Equal-width

[| | | |]
equal width

② Equal-frequency



③ Domain-specific

Age $\leq$ Infant
Salary

④ cluster-based

Group 4, Group 2, Group 3

Stage III

Category $\rightarrow$ Numeric

Gender

M F T

(Encoding)

one-hot

Categories

1	0	0
0	1	0
0	0	1

Label

0
1
2
3

ordinal

ordinal variable

Binary
Target

mean

0
1
2
3

00
01
10
11

Hash
entity